07 — Physics-Informed GPs and Boundary Conditions

> **What you'll learn:** how to put real physics (advection–diffusion / ocean currents) and exact boundary conditions (Dirichlet/Neumann/Robin) into a GP, via both the SPDE operator and linear-PDE-constrained kernels.
> **Prereqs:** 06 — the SPDE approach.
> **Why it matters here:** this is the concrete "incorporate physical processes and boundary conditions" ask; it underpins the future ocean-current-field model.

1. Two ways to make a GP obey physics

There are two complementary routes, both reducing to "constrain the GP with a **linear** differential operator":

- **A. The SPDE-operator route** (doc 06): change the operator on the left of `L X = ?` to encode the dynamics. The field's covariance then inherits the physics.
- **B. The constrained-kernel route:** start from a standard GP and **transform its kernel** so every sample exactly satisfies a linear PDE and/or boundary condition. If `f ~ GP(0, k)` and `g = ?f` for a linear operator ?, then g is also a GP with kernel ? ?′ k. So differential constraints map to kernel transformations — physics enters analytically.

Both work because **Gaussianity is preserved under linear operations** (differentiation, integration, linear PDE operators). Nonlinear physics needs approximation, but a lot of ocean transport is well captured by linear advection–diffusion.

2. The advection–diffusion SPDE (ocean currents)

The physical process most relevant to an AUV's drift is **advection–diffusion**: a quantity is **transported** by the current velocity field **and diffused** by mixing. The spatio-temporal SPDE (Clarotto, Allard, Romary, Desassis, 2022/2024) is:

```
∂X/∂t  +  (γ·∇)X  −  ∇·(D ∇X)  +  κ² X   =  ?(s, t)
        └─ advection ┘  └─ diffusion ┘   └ damping ┘   └ random forcing ┘
```

- **`γ`** — the **advection (drift) velocity** vector. Set it to the local current and the field's correlations **tilt and travel downstream** — features are carried along the flow, exactly like dye in a current. This is how the current itself enters the prior.
- **`D`** — the **diffusion tensor** (can be anisotropic): how fast structure smears out and in which directions (e.g. stronger horizontal than vertical mixing).
- **`κ²`** — damping/range term (as in doc 06).

- First-order `∂/∂t` makes it a genuinely **nonseparable spatio-temporal** model — space and time correlations are coupled, which separable kernels cannot represent.

Why this beats a plain space×time kernel

A naive spatio-temporal GP often uses a separable kernel `k_space · k_time`, which assumes the spatial pattern is frozen and just fades. Advection–diffusion produces **transported, evolving** correlations — physically correct for currents. That structure is the payoff of going physics-informed.

A numerical wrinkle: streamline diffusion

When advection dominates diffusion (fast current, little mixing), naive FEM discretization becomes unstable/oscillatory. The standard fix is **streamline-diffusion (SUPG) stabilization** — add a small amount of diffusion along the flow direction. Worth knowing because it's a required ingredient, not an optional tweak, in advection-dominated regimes (Clarotto et al. discuss exactly this).

3. Boundary conditions — making the GP respect the domain

A PDE needs boundary conditions to be well-posed, and they are how the GP learns about walls, the seabed, the surface, and known values. The three classical types:

Type	Constrains	Underwater example
Dirichlet	the **value** on the boundary: `X = g`	known current at an inflow boundary; zero anomaly at a reference
Neumann	the **normal derivative**: `∂X/∂n = h`	**no flow through the seabed** (zero normal velocity) — a hard physical fact
Robin	a mix: `aX + b ∂X/∂n = c`	partially absorbing / leaky boundaries

Two ways to enforce them:

- **In the SPDE/FEM mesh:** impose the condition on boundary nodes during assembly — exact and natural once you have a mesh. The choice of boundary condition genuinely changes the field near edges (a Neumann "no-flow" seabed makes the modelled current go along the bottom, not through it).
- **In the kernel (boundary-constrained GPs):** design mean and covariance so **every** sample satisfies the boundary condition by construction. Gulian et al. / Tan (JMLR v25, 23-1508) give closed-form **boundary-constrained kernels for Dirichlet, Neumann, Robin and mixed** conditions — e.g. expanding the kernel in the **Dirichlet eigenfunctions of the Laplacian** enforces zero-value boundaries globally and noiselessly. This route needs no mesh and is attractive on simple domains.

Why bother: enforcing physics-correct boundaries (a) removes nonsensical predictions near

edges,
(b) sharply reduces uncertainty there (the model "knows" the boundary), and (c) improves extrapolation into sparsely-observed regions — all valuable when bridging long DVL outages near the seabed.

4. Divergence-free / incompressibility constraints

Sea water is effectively **incompressible**: the current field should be **divergence-free** ($\nabla \cdot u = 0$). Using the constrained-kernel route (§1B) you can build a **vector-field kernel** whose samples are divergence-free by construction (curl-based / stream-function kernels). Physics-informed, boundary-constrained GP regression for fluid flow fields does exactly this (Jidling et al.; recent fluid-reconstruction work). For us this would make a reconstructed current field physically consistent, which improves drift prediction.

5. How this would serve the AUV (the payoff)

Put together, a physics-informed current model gives the navigator a prior over the water it is moving through:

- During **DVL bottom-lock loss**, the vehicle still measures **water-relative** velocity (or nothing);

a current-field GP predicts the **current**, so `ground velocity ≈ water-relative velocity + current` can be reconstructed → drift is bounded even without bottom-lock.

- Boundary conditions keep predictions sane near the seabed/surface.
- The SPDE/GMRF form (doc 06) keeps it computable, and combined

with Markov-in-time (doc 05) it can run as an efficient spatio-temporal filter.

6. Scope and honesty

This is the **most ambitious** thread and is **explicitly future work** (`implementation/04`). It needs: a meshing/FEM or constrained-kernel implementation, sparse linear algebra (`scipy`-class), an estimate of the current field (γ) from data or a model, and validation data. The first build deliberately stays with the simpler, high-value temporal velocity GP.

Key takeaways

- Physics enters a GP through a **linear operator** — either by editing the **SPDE operator** (doc 06)

or by **transforming the kernel** (`g=?f ⇒ k_g = ??′k`); Gaussianity survives linear operations.

- **Advection–diffusion** encodes ocean currents: γ (drift) tilts/transports correlations, `D`

(diffusion) smears them — a true nonseparable spatio-temporal model (needs streamline-diffusion stabilization when advection dominates).

- **Boundary conditions** (Dirichlet/Neumann/Robin) make the GP respect seabed/surface/walls exactly,

via the FEM mesh or **boundary-constrained kernels**; **divergence-free** kernels enforce incompressibility.

- Payoff: a physics-correct **current-field prior** that bounds drift during bottom-lock loss — future

work, but the principled end-state.

Further reading

- Clarotto et al. (2022/2024), *The SPDE approach for spatio-temporal datasets with advection and

diffusion*, arXiv:2208.14015.

- Gulian/Tan et al. (2024), Boundary-constrained Gaussian processes for ... linear PDEs, JMLR

v25/23-1508.

- Jidling et al., Linearly constrained Gaussian processes (divergence-free / curl-free kernels).
- Next: 08 — Cheap edge compute for GPs.